

Health Transformation Review (HTR) — Documentation Index

Platform version: 4.2.0 | **Last updated:** March 2026 | **Stack:** Next.js 16 · FastAPI · Supabase · Sanity · Groq

HTR (Health Transformation Review) is an intelligence and analytics platform for understanding, modeling, and navigating U.S. healthcare transformation. It covers five analytical pillars — Policy, Economics, Technology, Clinical, and Equity — through a continuously updated intelligence feed, AI-powered analysis, 19 interactive research tools, structured learning, and expert advisory services.

Documentation Map

Document	Audience	Description
User Guide	All users	Features, navigation, subscription tiers, quick-start
Developer Guide	Engineers	Local setup, env vars, API reference, adding features
Technical Architecture	Architects · Engineers	System design, data flow, service map, component map
Database Guide	Backend devs · DBA	Schema, RLS policies, migrations, Supabase client patterns
AI & RAG Guide	AI/ML engineers	RAG pipeline, LLM routing, ingest, embeddings, retrieval
Content Management	Editors · Content leads	Sanity Studio, 21 content types, editorial workflow
Research Lab	Analysts · Tool devs	19 tools across 6 labs, calculation reference, UI patterns
Operations Guide	DevOps · SRE	Deployment, monitoring, environment management, incident response
Academy & Coaching	Faculty · Product	Course structure, modules, tracks, personalized learning, coaching
Advisory Services	Advisory team · Sales	Service lines, client segmentation, five-pillar methodology, pricing
White Paper	Executives · Policymakers	Five-pillar framework, strategic rationale, health system context
HTI Methodology	Data scientists · Researchers	Health Transformation Index algorithm, data sources, formulas
Vermont Act 167	VT policymakers · Analysts	Act 167 analysis, AHEAD model, Vermont-specific implementation
Platform Improvement Plan	Leadership · Product	Roadmap: AI, architecture, database, security, compliance
Testing & Deployment	Engineers · Operators	End-to-end testing, staging, deployment runbook
Personalized Learning	Engineers · Product	Technical deep-dive on the AI curriculum generation feature
HTR Connect	Engineers · Product	HTR Connect community feature — architecture and implementation

Platform Overview

Five Pillars

Pillar	Color	Focus
Policy	Sky blue	Legislation, regulation, payment reform, state programs
Economics	Emerald	Value-based care, investment, market dynamics, cost analysis
Technology	Indigo	AI/ML, digital health, interoperability, EHR, FHIR
Clinical	Red	Hospital care, quality measures, workforce, delivery systems
Equity	Amber	SDOH, racial disparities, rural access, community health

Core Product Surfaces

Feature	Route	Description
Intelligence Feed	/	Curated real-time articles across all five pillars
Pillar Hubs	/ {pillar}	Dedicated analysis pages per pillar
State Dashboard	/ dashboard / [state]	Performance Index, RHT program data, hospital view
AI Analyst	/ chat	RAG-powered chat with the HTR knowledge base
Research Lab	/ research - lab	19 interactive analytical tools
HTI Dashboard	/ hti - dashboard	Real-time Health Transformation Index monitoring
Academy	/ academy	Courses, modules, tracks, webinars, personalized learning
Advisory Hub	/ advisory - hub	Expert consulting services and engagement tools

Technology Stack

Frontend:	Next.js 16 (App Router) · React 19 · TypeScript · Tailwind CSS v4
Backend:	FastAPI (Python) · LlamaIndex · Groq (Llama 3.3-70b) · OpenAI
Embeddings	
Database:	Supabase PostgreSQL · pgvector · Row Level Security
CMS:	Sanity v3 (21 content types) · GROQ queries · Portable Text
Auth:	Supabase Auth · JWT · 6-role RBAC (free → admin)
Payments:	Stripe Subscriptions (3 tiers: \$29/\$19/\$99/mo)
Email:	Loops.so (transactional) · Resend (system emails)
Hosting:	Vercel (frontend) · Railway (backend)
Monitoring:	Sentry (errors) · Vercel Analytics (performance)

Subscription Tiers

Tier	Price	Key Access
Free	\$0	Preview articles, state dashboard overview, limited AI queries
Subscriber	\$29/mo	Full articles, AI Analyst, Research Lab, Academy modules
Student	\$19/mo	Same as Subscriber (verified .edu email)
Professional	\$99/mo	Everything + Advisory Hub, priority AI model, API access
Advisory	Custom	Full platform + dedicated consulting engagement
Admin	Internal	Full platform access + admin dashboard

Repository Structure

Vermont-Health-Platform/	
└─ frontend/	# Next.js 16 application

├── app/	# App Router pages and API routes
│ ├── api/	# Next.js API routes (proxies to backend)
│ ├── admin/	# Admin dashboard (role-gated)
│ ├── academy/	# Learning management system
│ ├── chat/	# Full-page AI Analyst
│ ├── dashboard/	# State performance dashboard
│ ├── research-lab/	# 19-tool analytical suite
│ └── [pillar]/	# Economics, policy, technology, clinical, equity
├── components/	# Shared React components
│ ├── research/	# Individual research tool components (19 tools)
│ └── templates/	# PillarHub, HubPageTemplate, ArticlePageTemplate
├── lib/	# Data fetching, auth, Sanity client, utilities
└── public/	# Static assets
├── backend/	# FastAPI Python service
│ ├── routers/	# chat, ingest, api_v1, personalized_learning
│ ├── services/	# db, auth, llm, retrieval, indexing
│ └── data/	# Static data files and documentation
├── supabase/	# Database migrations and seeds
│ ├── migrations/	# SQL migration files
│ └── seed/	# Seed data scripts
└── docs/	# This documentation set

Quick Links

- **Production:** Deploy via Vercel (frontend) + Railway (backend)
- **CMS:** Sanity Studio at [\[frontend-url\]/studio](#) (admin role required)
- **Admin Dashboard:** [\[frontend-url\]/admin](#) (admin role required)
- **Health Check:** [\[backend-url\]/health](#)
- **API Docs:** [\[backend-url\]/docs](#) (FastAPI auto-generated OpenAPI)